Yuzheng Hu

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EDUCATION

University of Illinois at Urbana-Champaign — Urbana, IL, USA
Ph.D. in Computer Science — *Aug. 2021 – May. 2026*

- **Advisor**: Han Zhao
- **Thesis**: Advancing Generative AI via the Science of Data

Peking University — Beijing, China
B.S. in Mathematics (Math and Applied Math, Honor Track) — *Sep. 2017 – Jul. 2021*

- **Advisors**: Wei Lin and Liwei Wang

RESEARCH INTERESTS

I develop the **science of data**—a perspective that views data as an *optimizable*, *expandable* and *predictable* component of AI systems. To this end, I work on **data attribution**, **synthetic data**, and **privacy**.

PUBLICATIONS

⋆ stands for equal contribution or alphabetical order

[P1] A Survey of Data Attribution: Methods, Applications, and Evaluation in the Era of Generative AI
Junwei Deng⋆, **Yuzheng Hu⋆**, Pingbang Hu⋆, Ting-Wei Li⋆, Shixuan Liu⋆, Jiachen T. Wang, 21 others, Jiaqi W. Ma
Journal of Artificial Intelligence Research (**JAIR 2026**)

[P2] RL-IF: Tracking Non-local Training Data Influence for Online Reinforcement Learning
Shixuan Liu⋆, Cheng Tang⋆, **Yuzheng Hu⋆**, Fan Wu, Han Zhao, Jiaqi W. Ma
The 40th Annual Conference on Neural Information Processing Systems (**NeurIPS 2026**)

[P3] MAPLE: Metadata Augmented Private Language Evolution
Eli Chien⋆, **Yuzheng Hu⋆**, Ryan McKenna, Shanshan Wu, Zheng Xu, Peter Kairouz
The 3rd Conference on Language Modeling (**COLM 2026**)

[P4] ACTG-ARL: Differentially Private Conditional Text Generation with RL-Boosted Control
Yuzheng Hu, Ryan McKenna, Da Yu, Shanshan Wu, Han Zhao, Zheng Xu, Peter Kairouz
The 43rd International Conference on Machine Learning (**ICML 2026**)

[P5] OPUS: Towards Efficient and Principled Data Selection in Large Language Model Pre-training in Every Iteration 🤗
Shaobo Wang, Xuan Ouyang, Tianyi Xu, **Yuzheng Hu**, Jialin Liu, Guo Chen, Tianyu Zhang, Junhao Zheng, Kexin Yang, Xingzhang Ren, Dayiheng Liu, Linfeng Zhang
The 43rd International Conference on Machine Learning (**ICML 2026, Oral**)

[P6] A benchmark of expert-level academic questions to assess AI capabilities
Dataset contributor of Humanity's Last Exam; authored 4 of 2,500 expert-level questions, with one selected as a prize-winning entry (**top 550 worldwide**)
Nature (**2026**)

[P7] A Snapshot of Influence: A Local Data Attribution Framework for Online Reinforcement Learning
Yuzheng Hu⋆, Fan Wu⋆, Haotian Ye, David Forsyth, James Zou, Nan Jiang, Jiaqi W. Ma, Han Zhao
The 39th Annual Conference on Neural Information Processing Systems (**NeurIPS 2025, Oral**)

[P8] Taming Hyperparameter Sensitivity in Data Attribution: Practical Selection Without Costly Retraining
Weiyi Wang, Junwei Deng, **Yuzheng Hu**, Shiyuan Zhang, Xirui Jiang, Runting Zhang, Han Zhao, Jiaqi W. Ma
The 39th Annual Conference on Neural Information Processing Systems (**NeurIPS 2025**)

[P9] MergeBench: A Benchmark for Merging Domain-Specialized LLMs
Yifei He, Siqi Zeng, **Yuzheng Hu**, Rui Yang, Tong Zhang, Han Zhao
The 39th Annual Conference on Neural Information Processing Systems (**NeurIPS 2025, Datasets and Benchmarks Track**)

[P10] Empirical Privacy Variance
Yuzheng Hu*, Fan Wu*, Ruicheng Xian, Yuhang Liu, Lydia Zakynthinou, Pritish Kamath, Chiyuan Zhang, David Forsyth
The 42nd International Conference on Machine Learning (**ICML 2025**)

[P11] An Improved Autoregressive Evaluation Paradigm for Large Language Models
Jipeng Zhang*, Rui Pan*, **Yuzheng Hu***, Kashun Shum*, Guanyu Yao, Xiang Liu, Renjie Pi, Hanze Dong, Shizhe Diao, Yong Lin, Han Zhao, Tong Zhang
ACM Transactions on Intelligent Systems and Technology (**ACM TIST 2025**)

[P12] Most Influential Subset Selection: Challenges, Promises, and Beyond
Yuzheng Hu, Pingbang Hu, Han Zhao, Jiaqi W. Ma
The 38th Annual Conference on Neural Information Processing Systems (**NeurIPS 2024**)

[P13] Localize-and-Stitch: Efficient Model Merging via Sparse Task Arithmetic
Yifei He, **Yuzheng Hu**, Yong Lin, Tong Zhang, Han Zhao
Transactions on Machine Learning Research (**TMLR 2024**)

[P14] SoK: Privacy-Preserving Data Synthesis
Yuzheng Hu*, Fan Wu*, Qinbin Li, Yunhui Long, Gonzalo Munilla Garrido, Chang Ge, Bolin Ding, David Forsyth, Bo Li, Dawn Song
The 45th IEEE Symposium on Security and Privacy (**S&P 2024**)

[P15] HPL-ViT: A Unified Perception Framework for Heterogeneous Parallel LiDARs in V2V
Yuhang Liu, Boyi Sun, Yuke Li, **Yuzheng Hu**, Fei-Yue Wang
The 41st IEEE International Conference on Robotics and Automation (**ICRA 2024**)

[P16] Revisiting Scalarization in Multi-Task Learning: A Theoretical Perspective
Yuzheng Hu, Ruicheng Xian, Qilong Wu, Qiuling Fan, Lang Yin, Han Zhao
The 37th Annual Conference on Neural Information Processing Systems (**NeurIPS 2023**)

[P17] Understanding the Impact of Adversarial Robustness on Accuracy Disparity
Yuzheng Hu, Fan Wu, Hongyang Zhang, Han Zhao
The 40th International Conference on Machine Learning (**ICML 2023**)

[P18] Actor-critic is implicitly biased towards high entropy optimal policies
Yuzheng Hu*, Ziwei Ji*, Matus Telgarsky*
The 10th International Conference on Learning Representations (**ICLR 2022**)

[P19] Towards Understanding the Data Dependency of Mixup-style Training
Muthu Chidambaram, Xiang Wang, **Yuzheng Hu**, Chenwei Wu, Rong Ge
The 10th International Conference on Learning Representations (**ICLR 2022, Spotlight**)

[P20] Don't Waste Your Bits! Squeeze Activations and Gradients for Deep Neural Networks via TinyScript
Fangcheng Fu, **Yuzheng Hu**, Yihan He, Jiawei Jiang, Yingxia Shao, Ce Zhang, Bin Cui
The 37th International Conference on Machine Learning (**ICML 2020**)

Manuscripts

[M1] SHIP: Structured and Hierarchical Insight with Privacy
Kai Chen, Guisi Ni, Albert Cheu, Kassem Fawaz, **Yuzheng Hu**, Ryan McKenna, Peter Kairouz

[M2] A Unified Theory of Random Projection for Influence Functions
Pingbang Hu*, **Yuzheng Hu***, Jiaqi W. Ma*, Han Zhao*

Invited Talks

Differentially Private Conditional Text Generation with RL-Boosted Control

- Google Athena seminar — Sep. 2025
- Google ML Privacy seminar (Host: Natalia Ponomareva) — Sep. 2025

Empirical Privacy Variance

- Google ML Privacy seminar (Host: Matthew Jagielski) — Mar. 2025

Employment

Google Research — New York, NY, US
Research Scientist — *May. 2026 – now*

Internship Experience

Google Research — Mountain View, CA, US
Research Intern, Hosts: Zheng Xu, Ryan McKenna, Da Yu; Manager: Peter Kairouz — *May. 2025 – Nov. 2025*

- Developed a metadata-conditioned framework for DP synthetic text generation with [P4] and without [P3] fine-tuning

Jane Street Group, LLC — New York, NY, US
Quantitative Trading Intern — *May. 2024 – July. 2024*

- Ranked 2/200 (1/150 among interns) in the Tenth Annual Figgie World Championships
- Participated in mock trading games and worked on two desk projects (options and equities)

Damo Academy, Alibaba Group (U.S.) Inc. — Bellevue, WA, US
Research Intern, Mentor: Bolin Ding — *May. 2022 – Aug. 2022*

- Worked on the intersection of privacy and law
- Wrote a systematization of knowledge paper on privacy-preserving data synthesis [P14]

Research Experience

Simons Institute for the Theory of Computing — Berkeley, CA, US
Visiting Graduate Student, Program: Modern Paradigms in Generalization — *Aug. 2024 – Dec. 2024*

- Quantified the discrepancy between differential privacy and memorization in LLMs [P10], with Pritish Kamath and Chiyuan Zhang

University of Illinois at Urbana-Champaign — Urbaba, IL, US
Graduate Research Assistant, Advisor: Han Zhao — *Feb. 2022 – now*

- Thesis research: data attribution [P1] [P2] [P7] [P8] [P12] [M2], with Jiaqi Ma
- Other works: model merging [P9] [P13]; representational limitations of scalarization in multi-task learning [P16]; trade-off between robustness and fairness [P17]

University of Illinois at Urbana-Champaign — Urbaba, IL, US
Graduate Research Assistant, Advisor: Matus Telgarsky — *Aug. 2021 – Dec. 2021*

- Proved that actor-critic converges to high entropy optimal policy without explicit regularization [P18]

Duke University Remote
Undergraduate Research Assistant, Advisor: Rong Ge *Jun. 2020 – Feb. 2021*

- Demonstrated the data-dependency of mixup [P19]

School of Intelligence Science and Technology, Peking University Beijing, China
Undergraduate Research Assistant, Advisor: Liwei Wang *Aug. 2019 – Jun. 2020*

- Identified two non-local properties of the loss landscape of neural networks

School of Computer Science, Peking University Beijing, China
Undergraduate Research Assistant, Advisor: Bin Cui *Aug. 2018 – Aug. 2019*

- Introduced a distribution-aware, non-uniform quantization algorithm for multilayer perceptrons [P20]

Teaching Experience

UIUC CS 591 – Machine Learning Reading Group — Spring 2024, Fall 2024
UIUC CS 598 – Transfer Learning — Spring 2023
UIUC CS 498 – Trustworthy Machine Learning — Fall 2022

Services

Seminar Co-Organizer: UIUC Machine Learning Seminar — 2024 – 2026
Workshop Organizer and Primary Contact: DATA-FM @ ICLR 2026
Area Chair: NeurIPS 2026 Position Paper Track
Program Committee: TPDP (2026, 2025), ICLR SynthData (2025), ACL L2M2 (2025)
Reviewer: NeurIPS (2022-2025), ICML (2023-2026), ICLR (2024-2026), AISTATS (2024), TMLR

Honors and Awards

Top Area Chair: NeurIPS 2026 Position Paper Track — 2026
Top/Outstanding Reviewer: ICML 2026 (Top 25%), ICML 2025 (Top 2%), NeurIPS 2025 (Top 8%), NeurIPS 2023 (Top 10%)
ICML 2026 Oral (Top 0.7%, 168/23,918) — 2026
NeurIPS 2025 Oral (Top 0.3%, 77/21,575) — 2025
Saburo Muroga Endowed Fellowship (Top 4% in UIUC) — 2021-2022
Exceptional Award for Academic Innovation (Top 1% in Peking University) — 2020
Elite Undergraduate Training Program of Applied Math (Top 15% in SMS) — 2019-2021
Invited Participant, China TST for the 58th International Mathematical Olympiad (IMO) — 2017
Gold Medal, 32nd Chinese Mathematical Olympiad (CMO) (Top 60 in China) — 2016

Travel Grants

TPDP Student Travel Funding ($825) — 2025
IEEE S&P Student Travel Grants Award ($1,000) — 2024
NeurIPS 2023 Scholar Award ($2,000) — 2023

Skills

Programming Python, Shell Script, LaTeX, MATLAB
Frameworks & Packages PyTorch, HuggingFace